Parth Pundalik Pai

Curriculum Vitae

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Education

2022–26 **Indian Institute of Technology Bombay (IITB), Mumbai, India | GPA: 8.74/10**,
Bachelor of Technology in Mechanical Engineering,
Minor degree in Artificial Intelligence and Machine Learning

2025 **KTH Royal Institute of Technology, Stockholm, Sweden**,
Industrial Engineering and Management, Semester Exchange

Research Interests

Data Science, Machine Learning, Artificial Intelligence, Optimization, Music Information Retrieval

Research & Professional Experience

2025 **Associative Memory tasks for Adaptive LIF Neurons**
Research Assistant at KTH Stockholm, Sweden | Guide: Prof. Anders Lansner and Prof. Pawel Herman
- Implemented discrete leaky integrate-and-fire (LIF) spiking model, to mimic biological brain function
- Trained network on 350+ clear images, and tested recall on their distorted versions to reproduce clear patterns
- Assessed recall accuracy across n x n (n=10, 14, 16) discrete LIF networks vs continuous graded networks
- Quantified cutoff training patterns for 95% recall accuracy, revealing 28%, 25% and 40% deviations between LIF and continuous neuron models across 10×10, 14×14 and 16×16 network sizes

2025 **Machine Learning Intern**, *Jaguar Land Rover - TBSI (JLR), Bangalore, India*
Digital Twin: Neural Network based Surrogate Modelling in Vehicle Dynamics | Guide: Prof. N Vyas
- Developed parallelized Python pipeline using CarMaker APIs, reducing end-to-end simulation time by 96%
- Preprocessed data from 100k+ simulations generated using latin hypercube sampling for 27 input parameters
- Built Artificial neural networks to predict Roll, Understeer & Jacking with R^2 of 95% and RMSE/range < 0.05
- Implemented Optuna to infer vehicle input parameters for target metrics, accelerating design workflow by 75%

2023 **Model Predictive Control on CubeSat**, *IITB Student Satellite Program - (IITBSSP)*
Controls Researcher in the Technical Team | Guide: Prof. Varun Bhalerao
- Worked on 1U CubeSat satellite (10cm × 10cm × 10cm), focusing on altitude & position control development
- Designed Model Predictive Control (MPC) algorithm, stabilizing CubeSat attitude, reducing yaw rate by 10%
- Improved CubeSat positional stability by upgrading the control system from PID to receding horizon MPC

Course Projects

2024 **Machine Learning based analysis of external flow around Air-Foil**
Course: Applied Data Science and Machine Learning (ME228) | Guide: Prof. Alankar Alankar
- Built a computer-vision pipeline to extract and map 1M+ flow-field datapoints into high-res streamline plot
- Derived numerical streamline density & trained MLP achieving 93% accuracy in predicting pressure-coefficient
- Trained and tuned a Random Forest Regressor model across 10k+ air-foil configurations optimizing Camber value and angle-of attack attaining an R^2 score of 95%

2023 **Machine Learning Based Movie-Recommendation System**
Course: Staistical Machine Learning and Data Mining (ME781) | Guide: Prof. Asim Tewari
- Preprocessed a large dataset of 44k+ movies and applied AutoTokeniser to create compact description vector
- Built BERT based movie-description embeddings, comparing with prompt embeddings using cosine similarity
- Developed Gradio user interface enabling real-time prompt input and top 5 movie recommendation generation

2023 **Mountain Cargo Challenge using Arduino UNO**
Course: Introduction to Makerspace (MS101) | Guide: Prof. Ankit Jain and Prof. Dinesh Sharma
- Built a 4 wheel autonomous Line-follower robot climbing 30° inclines while transporting payload of 300 grams
- Integrated 3 IR sensors with Arduino UNO and achieved $> 95\%$ accuracy after calibration over 50+ test runs
- Appreciated as one of the best bots in MS101 course and awarded an AP grade for excpetional performance.

Scholastic Achievements

- 2024 Selected as the **sole** nominee from IIT Bombay to KTH Royal Institute of Technology for Semester Exchange
- 2023 Achieved Branch Change to Mechanical Engineering awarded to top **2%** in a cohort of 1400+ students
- 2023 Received an **AP** grade in the Makerspace101 course, achieved by only 7 individuals out of 600+ students
- 2022 Secured the **98th** percentile in **IIT-JEE Advanced** examination taken by nearly 150k+ candidates nationwide
- 2022 Achieved a **99.7** percentile score in the **IIT-JEE Mains** among 1 million candidates across the country

Technical Projects

2025 **Dashboard development for Battery Health monitoring** | JLR Global Hackathon
- Developed ML based Battery Health Monitoring system using NASA's Battery SOC data with 700k+ datapoints
- Trained LSTM & Random Forest models with R^2 of 90% estimating battery health from SOC time-series
- Developed dashboard for real-time battery monitoring & created an interactive UI using Gradio all in 36 hours

2024 **Music Generation using RNNs and LSTMs** | Season of Code 2024, Web and Coding Club, IIT Bombay
- Preprocessed 300+ Deutsch folk songs from ESAC dataset into 64-step time series sequences for model training
- Trained a 256-unit LSTM network on encoded sequence from Kern-format dataset to generate novel melodies
- Built custom inference pipeline using music21, decoding generated symbolic sequence into playable MIDI notes

2023 **Reinforcement learning for Atari Breakout** | Season of Code 2023, Web and Coding Club, IIT Bombay
- Built Atari Breakout environment using OpenAI Gym & processed 50k+ frames for agent-environment interaction
- Used ϵ-greedy exploration schedule from 1.0 $\rightarrow$ 0.1 over 100k steps for stable exploration-exploitation behaviour
- Implementented Frame Stacking to handle temporal dependencies and simplify the state-space complexity

Technical Skills

- Software: LaTeX, GitHub, Linux, Fusion 360, IPG CarMaker
- Languages: Python, C++, R, MATLAB, SQL, Arduino
- Packages: PyTorch, Transformers, Sckit-Learn, Numpy, Pandas, Matplotlib, Keras, HuggingFace

Relevant Coursework

- ML: Applied Data Science & Machine Learning, Statistical Machine Learning & Data Mining, Speech & Natural Language Processing & the Web, Intro to Machine Learning, Programming in Data Science
- Math: Linear Algebra, Differential Equations, Differential Calculus, Integral Calculus, Mathematical Methods in Engineering, Probability and Stochastic Processes I, Statistical Design of Experiments
- Others: Computer Programming & Utilization, Operations Modelling and Analysis, Introduction to Cryptography

Teaching and Mentorship

2024 **Undergraduate Teaching Assistant** | Course: Staistical ML and Data Mining | Guide: Prof. Asim Tewari
- Served as a TA for ME781, a 300 student course on Statistical ML & Data Mining by Mechanical Department
- Handled multiple student queries, providing timely clarification on ML concepts and evaluation criteria
- Assisted faculty with in-class quizzes, exam logistics, grading workflows and overall course administration

2024 **Project Mentor** | Season of Code 2024 & Summer of Science 2024, IIT Bombay
- Mentored a group of 10+ students in Season of Code on Predictive Modelling using Time-Series Forecasting
- Mentored 10+ students in Summer of Science on Large Language Models, teaching theory of LLMs
- Curated and Delivered 20+ high quality resources, conducting structured learning over an 8-week program

Extra Curriculars

Music
- Awarded Best Flautist – Special Mention for exceptional flute performance at Symphony's Battle of the Bands
- Won Gold Medal in Stageplay Music at 6th Inter-IIT Culturals Meet held at IIT Kharagpur, India
- Obtained Junior Hindustani Classical Music certification from Karnataka Board after 7 years of vocal training

Others
- Volunteered in the Versova Beach Cleaning program hosted by Abhyuday, the social body of IIT Bombay
- Mentored 100+ High-school students in Science & Mathematics as part of World Konkani Community Outreach
- Elected as First-Year Class Representatives, facilitating academic communication between department students